

Scalable Data Science
NPTEL 2026
Assignment-3

1. A document consists of the character sequence **adbdbadbcda**. Using contiguous substrings of length $k = 4$ (4-shingles), which of the following correctly represents the set of distinct 4-shingles extracted from the document?

- A. $\{adbd, abad, bcda\}$
- B. $\{aaaa, dddd, bbbb, dddd\}$
- C. $\{adb, dbd, bda, dab, aba, bad, dbc, bcd, cda\}$
- D. $\{adbd, dbda, bdab, daba, abad, badb, adbc, dbcd, bcda\}$

Solution Answer:(D) A 4-shingle is any contiguous substring of length 4, so the distinct 4-shingles are $\{adbd, dbda, bdab, daba, abad, badb, adbc, dbcd, bcda\}$.

2. Two users have liked 20 and 30 movies, respectively. Altogether, the number of distinct movies liked by at least one of them is 40. If the similarity between their preferences is measured using the Jaccard index, what is the Jaccard distance between the two users?

- A. 0.25
- B. 0.50
- C. 0.75
- D. 0.80

Solution Answer:(C) Since $|A \cap B| = 20 + 30 - 40 = 10$, the Jaccard distance is

$$1 - \frac{|A \cap B|}{|A \cup B|} = 1 - \frac{10}{40} = 0.75,$$

3. We have given a code for Count-Min sketch, but some lines are missing. Choose the right option to fill up the blank.

```
import numpy as np
class Count_Min_Sketch(object):
    def __init__(self, depth, width, hash_family, Mat=None):
        self.depth = depth
        self.width = width
        self.hash_family = hash_family
```

```

    if len(hash_family) != depth:
        raise ValueError("The number of hash functions must match the depth.")
    if Mat is None:
        self.Mat = np.zeros([depth, width], dtype=np.int32)
    else:
        self.Mat = Mat

    def add_elem(self, p, d):
        for i in range(self.depth):
            _____(blank)_____

```

- A. `self.Mat[i][self.hash_family[i](p) % self.width] = d`
- B. `self.Mat[i][self.hash_family[p](i) % self.width] += d`
- C. `self.Mat[i][self.hash_family[i](p) % self.width] += d`
- D. `self.Mat[i][self.hash_family[p](i) / self.width] = d`

Solution Answer:(C) `self.Mat[i][self.hash_family[i](p) % self.width] += d`

4. In the previous question if we add below function query an element in the class, which will be among the below option is right option to fill in the blank?

```

def query_elem(self, p):
    return ___(i.)___([_____(ii.)_____ % self.width] for i in range(self.depth))

```

- A. `(i.) min (ii.)self.Mat[i][self.hash_family[i](p)`
- B. `(i.) max (ii.)self.Mat[i][self.hash_family[p](i)`
- C. `(i.) min (ii.)self.Mat[i][self.hash_family[p](i)`
- D. `(i.) median (ii.)self.Mat[i][self.hash_family[i](p)`

Solution Answer:(A) `(i.) min (ii.)self.Mat[i][self.hash_family[i](p)`

5. Suppose a k -d tree of depth d has been built on a dataset of n points. Ignoring the cost of building the tree, what is the asymptotic time complexity of performing an exact membership query for a point?

- A. $O(d)$
- B. $O(n)$

C. $O(n + d)$

D. $O(nd)$

Solution Answer:(A) Exact search follows a single root-to-leaf path in the k -d tree, so the running time is proportional to the tree depth.

6. Suppose you have a dataset $\mathcal{D} \subseteq \mathbb{R}^n$, where distance is measured between any pair x and y from $\mathcal{D}$ is $\ell_2(x, y) = (\sum_i (x_i - y_i)^2)^{1/2}$. Your friend create a naive hash family $H = \{h_1, \dots, h_n\}$, where $h_i(x) = \lfloor x_i/w \rfloor$, where w is some parameter. Is this a LSH family for ℓ_2 ?

A. Not possible to say.

B. Yes.

C. No.

Solution Answer:(C)

7. In L_2 (Euclidean) LSH, a hash function is defined as:

$$h_{u,b}(x) = \left\lfloor \frac{u \cdot x + b}{w} \right\rfloor$$

where u is a random unit vector and $b \sim \text{Uniform}[0, w]$.

If $\|x - y\|_2 \gg 4w$, under what condition can a hash collision still occur?

(a) A. Random shift b forces collision.

(b) B. u is nearly orthogonal to $(x - y)$, making $|u \cdot (x - y)| < w$.

(c) C. u is aligned with $(x - y)$, maximizing projection.

(d) D. Buckets merge under scaling.

Solution Answer:(B) A collision requires:

$$|u \cdot (x - y)| < w$$

Even if $\|x - y\|$ is large, this can happen when u is nearly orthogonal to $(x - y)$, making the projected distance small.

8. You have a query q and want to find points such that distance to q is less than R . Suppose you trained a LSH data structure using a specific value of k and L . But for a specific query q' , suppose you look up the candidates returned by the data structure and find that none of the candidate points are within the desired distance bound R to the query q' . Which of the following changes would **NOT** help in getting candidates within the desired distance?

A. Add more tables (and corresponding hash functions) to the existing data structure and thereby increase set of candidates.

B. Augment the existing hash indices (i.e. increase k) by adding more hash functions in order to create each bin index.

C. For each hash table, inspect bins that differ from the bin allotted to q' by one bit.

- D. Decrease the number of random lines/hyperplanes, i.e. throw away some of the k hash functions for each hash table.

Solution Answer:(B)

9. A database stores two datasets in separate k - d trees T_1 and T_2 . Although T_1 contains more points than T_2 , both trees have identical depth d . Ignoring preprocessing costs, how do the complexities of exact membership queries compare?
- A. Queries on T_1 are asymptotically slower.
 - B. Queries on T_2 are asymptotically slower.
 - C. Queries on both trees have the same asymptotic complexity.
 - D. The complexity depends on the query point.

Solution Answer:(C) An exact membership query in a k - d tree follows a single root-to-leaf path. Since both T_1 and T_2 have the same depth d , the search complexity for each tree is $O(d)$.

10. A MinHash signature of length 100 is constructed for a document D using 100 independent hash functions. What does the entry in the 5th position of the signature represent?
- A. The hash value of the 5th shingle appearing in D .
 - B. The minimum hash value among all 100 hash functions applied to the shingles of D .
 - C. The minimum value obtained by applying the 5th hash function to all shingles present in D .
 - D. The 5th smallest hash value obtained by applying a single hash function to the shingles of D .

Solution Answer:(C) Each position of a MinHash signature corresponds to one hash function, so the 5th entry is the minimum value produced by the 5th hash function over all shingles of D .

11. In random hyperplane Locality-Sensitive Hashing (LSH), a vector v is assigned a bit according to

$$h(v) = \text{sign}(r \cdot v),$$

where r is a random vector. If two vectors v_1 and v_2 become increasingly aligned (i.e., the angle $\theta(v_1, v_2)$ between them decreases), how does the probability that they are assigned the same hash value change?

- A. The collision probability increases.
- B. The collision probability decreases.
- C. The collision probability remains constant.
- D. The collision probability converges to 0.5.

Solution Answer:(A) Since $P(h(v_1) = h(v_2)) = 1 - \theta(v_1, v_2)/\pi$, decreasing the angle between the vectors increases the collision probability.

12. You have a collection of sets $\{S_1, S_2, \dots, S_m\}$ each of size n . Suppose for each set S_i , you define a hash of the set by just picking an element of the set uniformly at random, $h(S_i)$ is a uniformly chosen element of S_i . Will the $\mathbb{P}[h(S_i) = h(S_j)]$ for two sets S_i and S_j be equal to the Jaccard Similarity.
- A. No. Equal to size of smaller set divided by larger set the set
 - B. Yes. Equal to the Jaccard Similarity of two sets.
 - C. No. Can be independent of the Jaccard Similarity
 - D. Cannot say without knowledge of set elements

Solution Answer:(C) No. can be independent of the Jaccard Similarity.
Consider that each set has n elements. In this case $\mathbb{P}[h(S_i) = h(S_j)] = \frac{1}{n}$

13. For a dataset of Boolean vectors in d dimensions and Hamming distance how many functions are there in the LSH family.
- A. 2^d
 - B. d
 - C. $2d$
 - D. $d/2$

Solutino Answer:(B) LSH family for Hamming distance is constructed using a set of functions $f_i(x)$ which denotes the i^{th} coordinate (bit) of x . And for d -dimensional vectors the various coordinates (bits) that we can have is at most d .

14. For cosine (angle-based) similarity over vectors $x, y \in \mathbb{R}^d$, a standard LSH family uses random hyperplanes:

$$h_r(x) = \text{sign}(r \cdot x)$$

where r is a random unit vector.

The collision probability is:

$$P(h_r(x) = h_r(y)) = 1 - \frac{\theta}{\pi}$$

where θ is the angle between x and y .

If $\theta \rightarrow \frac{\pi}{2}$, what happens?

- (a) A. Collision probability $\rightarrow 0.5$
- (b) B. Collision probability $\rightarrow 0$
- (c) C. Projection becomes undefined
- (d) D. Hash switches from AND to OR behavior

Solution Answer:(A) When $\theta = \frac{\pi}{2}$,

$$P(h_r(x) = h_r(y)) = 1 - \frac{\theta}{\pi} = 1 - \frac{\pi/2}{\pi} = 1 - \frac{1}{2} = \frac{1}{2}$$

Thus, orthogonal vectors collide with probability 0.5, meaning the hash behaves like a fair coin flip at this boundary and loses discriminative power.

15. You have a stream of numbers and you want to get the approximate frequency of elements present in your stream. So, you implement Count sketch. But you make a “small” mistake while implementing it. The random function g which was supposed to be defined as $g : [n] \rightarrow \{-1, 1\}$, you mistakenly define it as $g : [n] \rightarrow \{-1/3, 1/3\}$. Your estimate for a still uses the same formulae as before. What do you think will be the expected value of the estimates you get for any element a i.e. $\mathbb{E}[f_a]$?

- A. $\mathbb{E}[f_a] = f_a$
- B. $\mathbb{E}[f_a] = f_a/6$
- C. $\mathbb{E}[f_a] = f_a/3$
- D. $\mathbb{E}[f_a] = f_a/9$

Solution Answer:(D) $\mathbb{E}[f_a] = f_a/9$

Here in the expectation calculation $g(a)^2$ will contribute $\frac{1}{9}f(a)$.

16. Suppose we have a (d_1, d_2, p_1, p_2) -sensitive family $\mathcal{F}$. We construct a new family $\mathcal{G}$ as follows. Each function $g \in \mathcal{G}$ is formed from a set of r independently chosen functions of $\mathcal{F}$, say $f_1, f_2, \dots, f_r$ for some fixed value of r . Now, $g(x) = g(y)$ if and only if for all $i = 1, \dots, r$, $f_i(x) = f_i(y)$. Which of the following statement is correct?

- A. $\mathcal{G}$ is an (d_1, d_2, p_1, p_2) family
- B. $\mathcal{G}$ is an (d_2, d_1, p_2, p_1) family
- C. $\mathcal{G}$ is an (d_1, d_2, p_1^r, p_2^r) family
- D. $\mathcal{G}$ is an $(d_1, d_2, (1 - p_1)^r, (1 - p_2)^r)$ family

Solution Answer:(C)

17. A distributed system consists of n machines, each observing its own data stream and maintaining either a Count-Min sketch or a Count sketch using the same parameters and hash functions. If the goal is to estimate statistics over the union of all streams, which of the following is true?

- A. The sketches from different machines can be combined to obtain a sketch of the aggregated stream.
- B. The sketches cannot be combined; all raw data must be collected at a central server.
- C. Whether the sketches can be combined depends on the distribution of the streams.

Solution Answer:(A) Count-Min sketches and Count sketches are linear sketches, so sketches from multiple streams can be added entry-wise to obtain the sketch of the combined stream.

18. In Count Sketch, each row uses a hash function $h_i : U \rightarrow \{1, \dots, d\}$ and a sign hash function $g_i : U \rightarrow \{-1, +1\}$. While computing

$$E[Y_i] = E[g_i(x) \cdot A[i, h_i(x)]],$$

the expected contribution of all collision terms $y \neq x$ becomes 0.

What property of g_i ensures this cancellation?

- (a) A. g_i forces all frequencies to be positive.
- (b) B. For $x \neq y$, $g_i(x)$ and $g_i(y)$ are independent with $E[g_i(y)] = 0$.
- (c) C. g_i is 4-wise independent and scales quadratically.
- (d) D. g_i converts L_1 norm into L_2 norm.

Solution Answer:(B) For collision term with $y \neq x$,

$$E[g_i(y)] = 0 \quad \text{and} \quad g_i(x), g_i(y) \text{ are independent}$$

So,

$$E[g_i(x) \cdot g_i(y)] = E[g_i(x)]E[g_i(y)] = 0$$

Hence all cross terms cancel.

19. Recall the *Band Construction* technique which we used to amplify the probabilities. In that we used k hash functions inside a band and L hash tables (i.e. L rows of bands). The probability that two points x and y collide is given as $1 - (1 - p^k)^L$. Let us call this as k -way *AND* followed by L -way *OR*. Now what would the probability of collision be if we do k -way *OR* followed by L -way *AND*.

- A. It doesn't matter, the probability remains same i.e. $1 - (1 - p^k)^L$
- B. The probability changes to $(1 - (1 - p^k))^L$
- C. The probability changes to $(1 - (1 - p)^L)^k$
- D. The probability changes to $(1 - (1 - p)^k)^L$

Solution Answer:(D)

20. Suppose a data stream has a highly skewed frequency distribution where a few elements are extremely dominant. To configure a Count Sketch data structure that guarantees an error bound within $\pm \epsilon \|f\|_2$ with failure probability at most δ , how should the internal matrix dimensions—number of rows (w) and number of columns per row (d)—be chosen?

- (a) A. $w = \log \frac{1}{\delta}$ and $d = \frac{3}{\epsilon^2}$
- (b) B. $w = \frac{3}{\epsilon^2}$ and $d = \log \frac{1}{\delta}$
- (c) C. $w = \log \frac{1}{\delta}$ and $d = \frac{2}{\epsilon}$
- (d) D. $w = \frac{2}{\epsilon^2}$ and $d = \log \frac{1}{\delta}$

Solution Answer: (A) In Count Sketch, variance of each estimator is $O(\|f\|_2^2/d)$. By Chebyshev, to get error $\pm \epsilon \|f\|_2$, we need:

$$d = \Theta\left(\frac{1}{\epsilon^2}\right) \Rightarrow d = \frac{3}{\epsilon^2}$$

Each row succeeds with constant probability, so boosting success to $1 - \delta$ uses median trick:

$$w = \Theta\left(\log \frac{1}{\delta}\right)$$